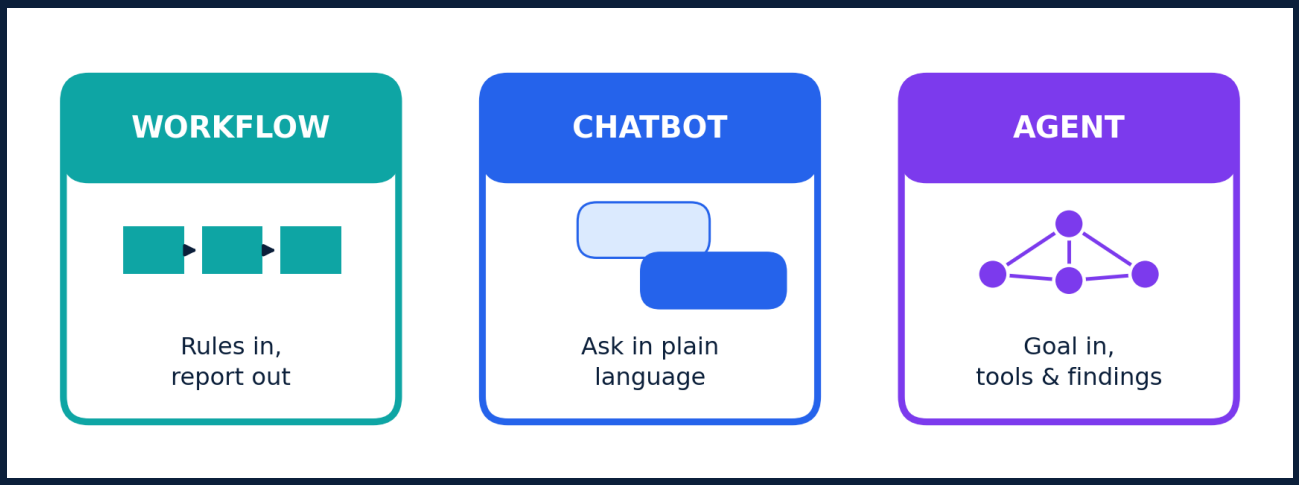
LEARN WITH RUPAYAN • PROJECT INTELLIGENCE DEMO

AI Decision Demo

Workflow vs AI Chatbot vs AI Agent

Comparing deterministic automation, conversational AI and goal-oriented agents on one business problem.



CORE IDEA

Choose the simplest approach that reliably solves the business problem. Add autonomy only when it creates measurable value.

Project Overview

A practical project-intelligence demo from **Learn with Rupayan** comparing three approaches to a business problem: deterministic workflow automation, conversational AI, and goal-oriented AI agents.

CORE IDEA

Choose the simplest approach that reliably solves the business problem. Add autonomy only when it creates measurable value.

The demo uses a project-management scenario to compare the three approaches. They are **not mutually exclusive**: an enterprise solution can combine all three.

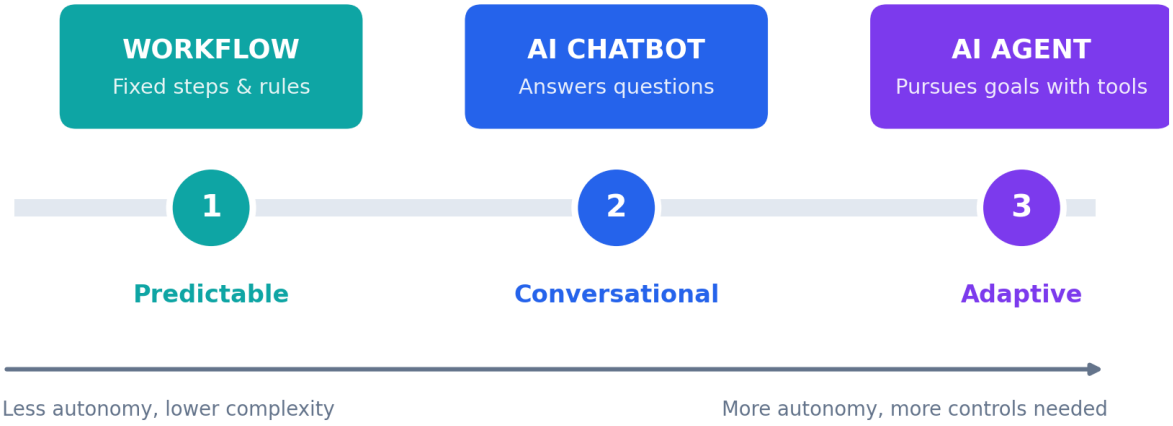


Figure 1. The three approaches along the autonomy spectrum.

1 Workflow Automation
Executes predefined rules and steps to produce predictable outputs.

2 AI Chatbot
Answers natural-language questions using project data and, where configured, retrieved document evidence.

3 AI Agent
Receives a broader goal and uses tools to investigate information and prepare findings or recommendations.

Example business question

EXAMPLE QUESTION

Review the project portfolio, identify risks and upcoming milestones, and highlight what requires management attention.

What the Demo Does

Depending on the selected mode, the application can:

- Read structured project, risk, and milestone data.
- Apply predefined rules to produce repeatable results.
- Answer questions about project status, risks, milestones, and supporting evidence.
- Retrieve relevant passages from an indexed document collection when retrieval is configured.
- Use tools to gather evidence for a broader investigation.
- Produce a project summary or executive-style investigation report.

Note: This is an educational demo, not a production project-management system. Validate its outputs before using them for real business decisions.

1 Workflow Automation

Predictable, repeatable, rule-based

Definition: A workflow executes a known sequence of steps and rules. Its logic is mostly defined in advance.

TYPICAL FLOW Input data → Predefined rules → Processing → Report or action

Example: Read project CSV files, check project status and milestone conditions, flag records that match predefined rules, and produce a report.

ADVANTAGES

- ✓ Predictable and repeatable.
- ✓ Usually straightforward to test and audit.
- ✓ Often efficient for stable, rule-based tasks.
- ✓ Can work without an LLM when ordinary code is sufficient.

LIMITATIONS

- ✗ Does not inherently interpret open-ended goals.
- ✗ New business rules may require code or configuration changes.
- ✗ Output quality depends on input data and rule correctness.

BEST SUITED FOR Scheduled reporting, data validation, notifications, routine checks, and other well-defined tasks.

2 AI Chatbot

Natural-language questions with optional retrieval

Definition: A chatbot accepts questions in natural language and generates conversational answers. Depending on its design, it may use model knowledge, supplied context, connected data, or a retrieval system.

TYPICAL FLOW User question → Retrieve relevant information → Add context to prompt → LLM generates answer

Example: A user asks, “Which projects are at risk, and what are their main blockers?” The chatbot uses available project data and retrieved evidence to produce a summary.

ADVANTAGES

- ✓ Users can ask questions in their own words.
- ✓ Supports exploration, summaries, explanations, and follow-up questions.
- ✓ Retrieval-Augmented Generation (RAG) can ground answers in relevant internal documents.

LIMITATIONS

- ✗ May retrieve incomplete or irrelevant evidence.
- ✗ May generate incorrect or unsupported statements.
- ✗ A citation is useful only if it supports the claim.
- ✗ Does not automatically plan or execute a multi-step investigation.

BEST SUITED FOR Question answering, knowledge access, document summarization, and conversational exploration.

3 AI Agent

Goal-oriented, tool-using, multi-step

Definition: An AI Agent works toward a goal and can select tools or steps dynamically within configured boundaries. Depending on its design, it may plan, call tools, inspect results, and decide what to do next.

TYPICAL FLOW Business goal → Select a step/tool → Execute tool → Inspect result → Continue or finish → Report

Example: The user asks the agent to investigate the portfolio. It reads structured records, searches indexed documents when available, combines evidence, and generates findings and recommendations.

ADVANTAGES

- ✓ Can handle multi-step tasks not fully specified as a fixed sequence.
- ✓ Can combine information from multiple tools or sources.
- ✓ Can adapt the next step to intermediate results.
- ✓ Can create a consolidated report from a broader goal.

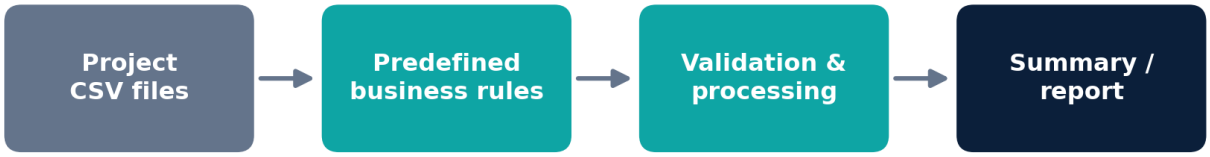
LIMITATIONS

- ✗ More complex to build, test, monitor, and maintain.
- ✗ May choose an unsuitable tool or miss important evidence.
- ✗ Multiple model calls and tool executions may increase latency and cost.
- ✗ A polished final report does not prove every intermediate step was correct.
- ✗ Not necessarily fully autonomous; autonomy depends on design and permissions.

BEST SUITED FOR Goal-oriented investigations and multi-source analysis where dynamic tool use justifies the additional complexity.

High-Level Architecture

Workflow



Every step and condition is defined in advance

Figure 2. The steps and conditions are specified in advance. Ordinary code may be enough for deterministic tasks.

AI Chatbot with Retrieval (RAG)

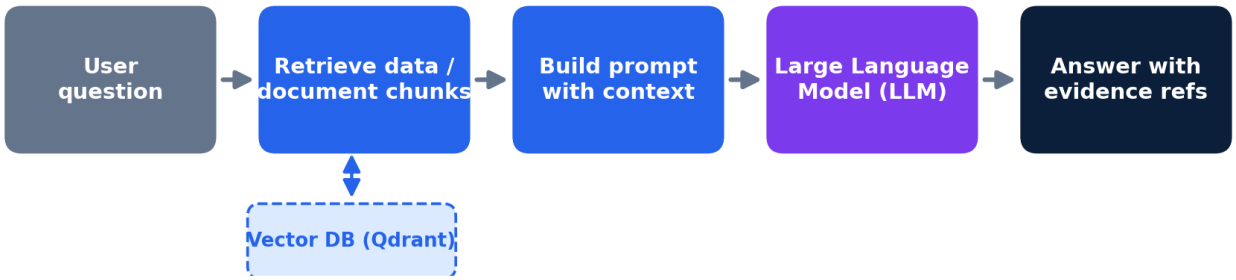


Figure 3. A retrieval pipeline commonly includes document ingestion, chunking, embeddings, vector indexing, retrieval, and prompt construction.

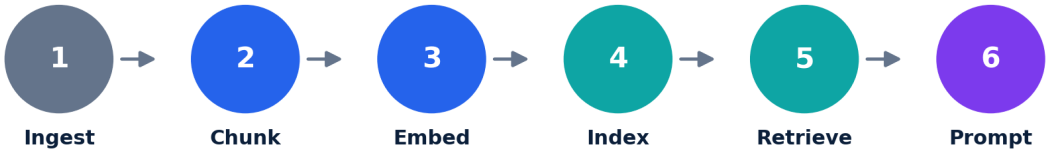


Figure 4. Typical stages of a retrieval pipeline.

AI Agent with Tool Calling



Figure 5. The application, not the LLM alone, should enforce permissions, validate arguments, execute tools, and control allowed actions.

Key Concepts

Large Language Model (LLM)	A model that processes and generates language. In this demo, it may interpret questions, create a search query, or synthesize supplied evidence into a report. Its output is not automatically factual and should be checked against evidence.
Prompt	Instructions and context supplied to a model. Prompts can define the goal, evidence, output structure, and rules such as “do not invent missing facts.” Prompt instructions are not a replacement for programmatic security controls.
Embeddings	Numeric representations of text used to compare semantic similarity. They help retrieve document passages related to a question even when the wording differs.
Vector Database	A system that stores vectors and supports similarity search. In this demo’s configured retrieval architecture, **Qdrant** is used for vector retrieval. A vector database does not guarantee that results are correct, complete, current, or authorized.
Retrieval-Augmented Generation (RAG)	RAG combines retrieval with generation: (1) retrieve potentially relevant information, (2) supply it to the LLM, (3) generate an answer based on the supplied context. RAG can improve grounding but does not eliminate hallucinations or retrieval failures.
Tool Calling	Lets a model request a specific function or operation, such as reading project records or searching documents. The application executes the function and returns its result to the model. **Tool calling is not the same as autonomous execution.** Validate tool requests in the application, and give tools only the permissions they need.
Human-in-the-Loop	A person reviews or approves an output or action at a defined point. Particularly important for financial, legal, security-sensitive, or otherwise high-impact operations.
Observability and Evaluation	**Observability:** logs, traces, tool results, latency, errors, and cost that help operators understand system behavior. **Evaluation:** tests that measure whether retrieval, answers, tool selection, and final outputs meet defined quality criteria. Both are important for reliable enterprise deployment.

Tool Calling: How It Works

A typical tool-calling cycle:

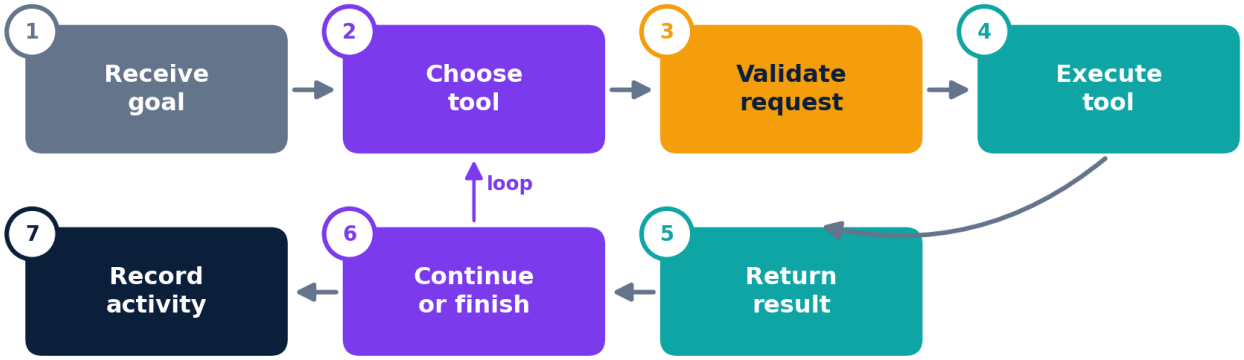


Figure 6. The seven-step tool-calling cycle.

1. **Receive the goal:** The system receives a user question or business objective.
2. **Choose a tool:** The LLM may request a tool such as `get_project_portfolio`, `get_open_risks`, `get_milestones`, or `search_project_documents`.
3. **Validate the request:** The application checks the tool name, arguments, permissions, and limits.
4. **Execute the tool:** The application runs the relevant function.
5. **Return the result:** The tool result is sent back to the model as data.
6. **Continue or finish:** The model may request another tool or produce a final answer.
7. **Record the activity:** Logs capture tool calls, errors, and relevant execution metadata.

SECURITY PRINCIPLE

The model should not have direct, unrestricted access to a database, filesystem, shell, or business system. Expose narrowly scoped functions and enforce authorization in the application.

Tools in this project's agent design

Tool	Purpose
`get_project_portfolio`	Read the structured project portfolio
`get_open_risks`	Read open-risk records
`get_milestones`	Read milestone records
`search_project_documents`	Retrieve relevant evidence from the indexed document collection

The exact execution path depends on the application version. A logged tool operation is not necessarily one LLM request: one model response can request a tool, and the application may execute several local tools during an investigation.

Advantages and Limitations: Quick Comparison

Dimension	Workflow	AI Chatbot	AI Agent
Main interaction	Predefined trigger/process	User question	Business goal
Execution	Fixed steps and rules	Answer generation; may use retrieval	May select tools and steps dynamically
Flexibility	Low to medium	Medium to high for questions	Potentially high for multi-step tasks
Predictability	Usually high for deterministic logic	Model output can vary	Tool choices and intermediate steps can vary
Typical cost pattern	Often low for deterministic processing	Model and possible retrieval costs	Potentially higher due to multiple calls and tools
Testing focus	Rules, inputs, edge cases	Retrieval quality and answer correctness	Tool choice, intermediate steps, final output, failure handling
Main risk	Incorrect rules or data	Unsupported or incorrect answers	Incorrect decisions or tool use across steps
Good fit	Repetitive, defined tasks	Information access and Q&A	Goal-driven, multi-step investigation

These are general tendencies, not universal laws. Actual cost, effort, and reliability depend on the use case, implementation, model, data, and operating environment.

Illustrative profile of general tendencies

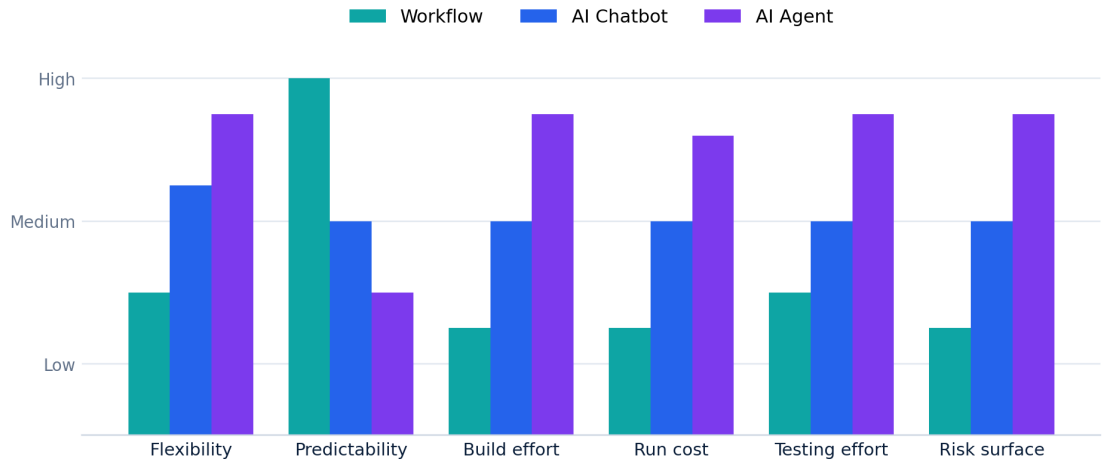


Figure 7. Qualitative, illustrative view only. Not measured data.

Risks, Security, and Governance

Common risks

Incorrect output
The system may produce wrong or incomplete results.

Poor data quality
Missing, stale, duplicated, or conflicting data can mislead any approach.

Hallucination
An LLM may generate unsupported statements.

Prompt injection
Malicious instructions in user input or retrieved content may influence model behavior.

Sensitive data exposure
The system may disclose information a user or tool should not access.

Over-reliance
Users may trust a polished answer without checking evidence.

Operational failure
API outages, rate limits, timeouts, and tool errors may interrupt execution.

Controls to consider

For all approaches

- Authenticate users and enforce authorization.
- Minimize sensitive data and apply retention policies.
- Validate inputs and outputs.
- Monitor failures, latency, and operating cost.
- Keep appropriate audit logs.
- Define owners, escalation paths, and incident-response procedures.

Additional controls for chatbots and RAG

- Enforce document permissions during retrieval.
- Track source documents and retrieval results.
- Test for unsupported claims and misleading citations.
- Defend against prompt injection and data leakage.
- Evaluate retrieval relevance and answer quality.

Additional controls for agents

- Use least-privilege permissions.
- Allowlist tools and validate tool arguments.
- Set maximum steps, timeouts, and cost limits.
- Separate read-only tools from tools that change data or trigger external actions.
- Require human approval for sensitive or irreversible actions.
- Log tool requests, results, errors, and approvals.
- Provide a way to stop execution and recover from failures.

Governance

Governance questions

- Who owns the system and approves changes?
- What data can it access?
- What actions is it allowed to perform?
- Which outputs require human review?
- How will quality, cost, and security be measured?
- How are errors, incidents, and user feedback handled?



Figure 8. Autonomy and risk profile (illustrative).

PRINCIPLE

More autonomy generally requires stronger controls. A read-only investigation agent has a different risk profile from an agent authorized to update records or execute transactions.

Cost, Time, and Engineering Effort

There is no single universal price for a workflow, chatbot, or agent.

Cost or effort area	Workflow	AI Chatbot	AI Agent
Initial build	Often low for simple rules	Often medium	Often medium to high
Model usage	Optional	Common	May involve multiple calls
Retrieval infrastructure	Optional	Needed for document-grounded RAG	Needed if the agent searches documents
Testing	Rules and edge cases	Retrieval and answer quality	Tools, intermediate decisions, and final results
Operations	Scheduling and integration monitoring	Model, retrieval, and answer monitoring	Model, tool, orchestration, security, and cost monitoring

This table is qualitative. A complex workflow can require more effort than a simple chatbot. Actual costs depend on model pricing, usage volume, hosting, platform licences, integrations, security requirements, and maintenance. Compare total cost of ownership and measurable outcomes, not just model-token cost.

Enterprise Implementation Guidance

1	**Define the business problem.** Specify the user, task, desired outcome, and acceptable failure rate.
2	**Start with the simplest suitable design.** Use a workflow for deterministic logic, a chatbot for conversational access, or an agent when dynamic multi-step work is justified.
3	**Prepare and govern the data.** Identify approved sources, data owners, permissions, freshness, and retention requirements.
4	**Build a narrow prototype.** Keep scope and permissions limited.
5	**Evaluate with realistic test cases.** Include normal cases, edge cases, missing data, conflicting evidence, and failure scenarios.
6	**Add security and observability.** Implement access controls, logging, monitoring, cost limits, and incident handling.
7	**Use human approval where appropriate.** Do not delegate sensitive decisions or irreversible actions without suitable controls.
8	**Measure business value.** Track accuracy, time saved, cost per task, user outcomes, failure rates, and maintenance effort.
9	**Expand gradually.** Increase scope or autonomy only when evaluation shows the additional complexity is worthwhile.

Choosing an approach

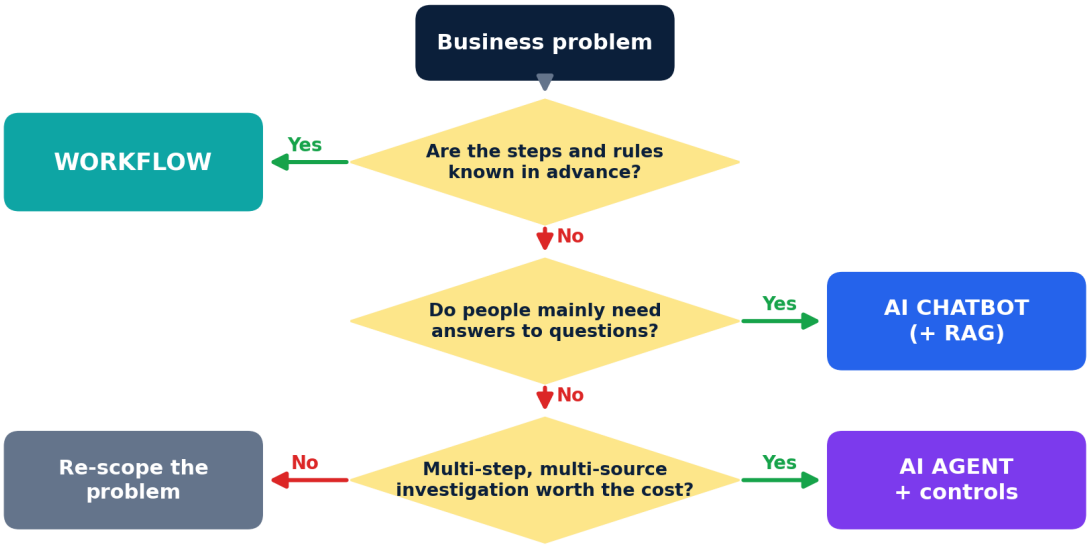


Figure 9. A simplified decision guide: start simple and add autonomy only when justified.

How to Evaluate the Demo

Use the same business problem across all three approaches:

TEST PROMPT

“Summarize the project portfolio, identify open risks and upcoming milestones, and highlight what requires management attention.”

Compare:

- [ ] Did the workflow apply its rules correctly?
- [ ] Did the chatbot answer using relevant evidence?
- [ ] Did the agent use appropriate tools and cover the required sources?
- [ ] Are project IDs, dates, and risk references supported by source records?
- [ ] Did the system distinguish missing evidence from evidence that no risk exists?
- [ ] Are recommendations proportionate to the findings?
- [ ] Were errors or source conflicts made visible?
- [ ] Was the output produced at acceptable cost and time?

Do not judge quality only by how polished the final report looks. Verify important claims against source records.

Important Notes

- This is an educational demo, not a production-ready enterprise control system.
- Actual capabilities depend on the configured code, data, model providers, and retrieval index.
- Document retrieval can return no results or irrelevant results; a vector database does not guarantee good evidence.
- Tool calling does not automatically mean an agent is autonomous or safe.
- Review the final report before using it for real management decisions.
- Vendor prices, model availability, and licensing change over time; verify current terms before production planning.

FINAL TAKEAWAY

Workflow for predictable execution.
Chatbot for conversational information access.
AI Agent for goal-driven, multi-step investigation, when the added autonomy is worth the cost and risk.

Choose the simplest approach that reliably solves the business problem. Add autonomy only when it creates measurable value.

Created for the Learn with Rupayan learning series: “Do You Really Need an AI Agent? — Workflow vs AI Chatbot vs AI Agent.”